

Quiz (Carolina Reaper)

- Q1.** Solve the system: $x^2 + y^2 = 25$, $y = x^2 - 4$. What is the solution set?
(A) (3, 5), (-3, 5)
(B) (-2, 0), (2, 0)
(C) (0, 5), (0, -5)
(D) (1, -3), (-1, -3)
(E) (4, 3), (-4, -3)
- Q2.** A logistics company uses the equations $d = 2t^2 + 5t + 1$ and $d = 3t^2 - 2t - 1$ to model the distance traveled by two delivery trucks. What is the time at which they are equidistant from the warehouse?
(A) $t = 1$ hour
(B) $t = 2$ hours
(C) $t = 3$ hours
(D) $t = 4$ hours
(E) $t = 5$ hours
- Q3.** Graph the system: $y = x^2 - 4x + 3$, $y = -x^2 + 2x + 1$. What is the number of intersection points?
(A) 0
(B) 1
(C) 2
(D) 3
(E) 4
- Q4.** The equations $p = 2x^2 + 3y^2$ and $p = x^2 - 2y^2 + 1$ model the pressure at different points in a pipeline. What is the solution to the system?
(A) (1, 1)
(B) (2, 0)
(C) (0, 1)
(D) (1, 0)
- (E) (0, 0)
- Q5.** A shipping company uses the equations $c = 2x^2 + y^2$ and $c = x^2 + 2y^2$ to model the cost of transporting goods. What is the relationship between x and y ?
(A) $x = y$
(B) $x = -y$
(C) $x^2 = y^2$
(D) $x = 2y$
(E) $y = 2x$
- Q6.** The system $x^2 + y^2 = 16$, $x^2 - y^2 = 4$ has how many real solutions?
(A) 0
(B) 1
(C) 2
(D) 4
(E) 8
- Q7.** The traffic flow equations $f = 3x^2 - 2y^2$ and $f = 2x^2 + y^2$ have a solution set of?
(A) (0, 0), (1, 1)
(B) (0, 0), (-1, 1)
(C) (1, -1), (-1, 1)
(D) (0, 1), (0, -1)
(E) (1, 0), (-1, 0)
- Q8.** The system $y = x^2 + 2$, $y = -x^2 - 2$ has what type of solution set?
(A) No solution
(B) One solution
(C) Two solutions
(D) Infinite solutions
(E) dependent

Open Ended Questions (FRQ)

- Q9.** Solve the system $x^2 + y^2 = 20$, $y = x^2 - 5$ and interpret the results in the context of transportation logistics.
- Q10.** A delivery company uses the equations $t = x^2 + 2y^2$ and $t = 2x^2 - y^2$ to model the time taken to deliver packages. Find the values of x and y that minimize the delivery time and explain the practical implications of your solution.

Chef's Tips

- Encourage students to use graphical methods to visualize the solutions.
- Emphasize the importance of checking the solutions in the original equations.
- Remind students to consider the practical context of the problems when interpreting the results.